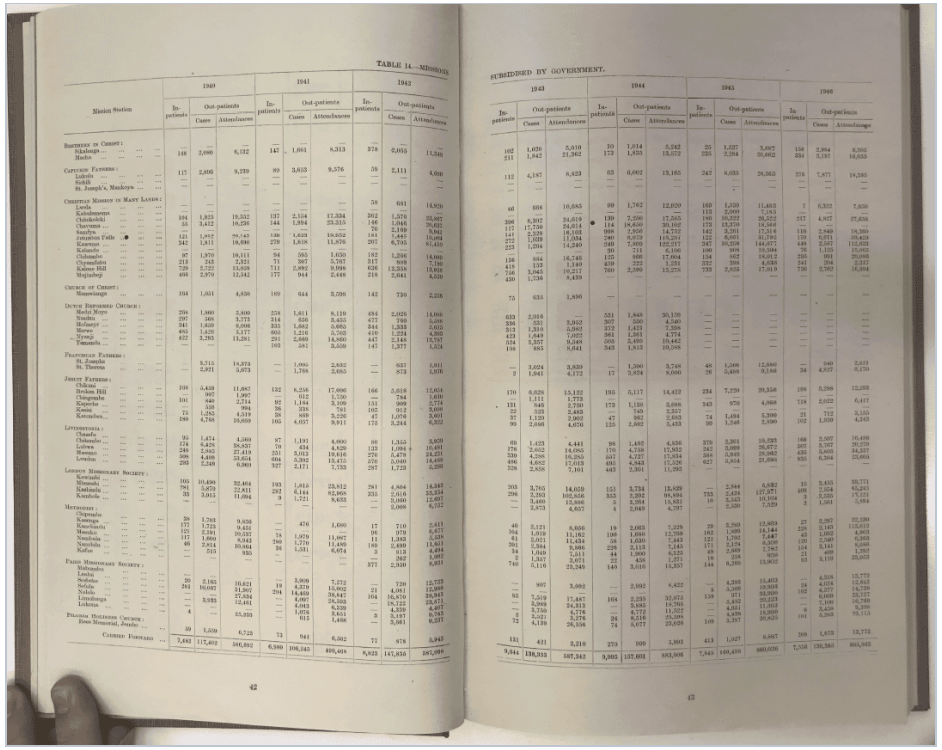
# The Long Shadow of Mission Medicine

Colonial missionary healthcare, institutional trust, and the spatial burden of disease in Sub-Saharan Africa

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Stellenbosch University, Laboratory for the Economics of Africa's Past (LEAP)

Wellcome Early-Career Award | 355277/Z/26/Z | January 2027



Northern Rhodesia medical report, Table 14: mission stations subsidised by government, 1940 to 1946

# Clinics are within reach, but many people don't use them

## The usual explanations

Poverty

Distance

Facility quality

↓

…leave a gap: similar communities still differ in use

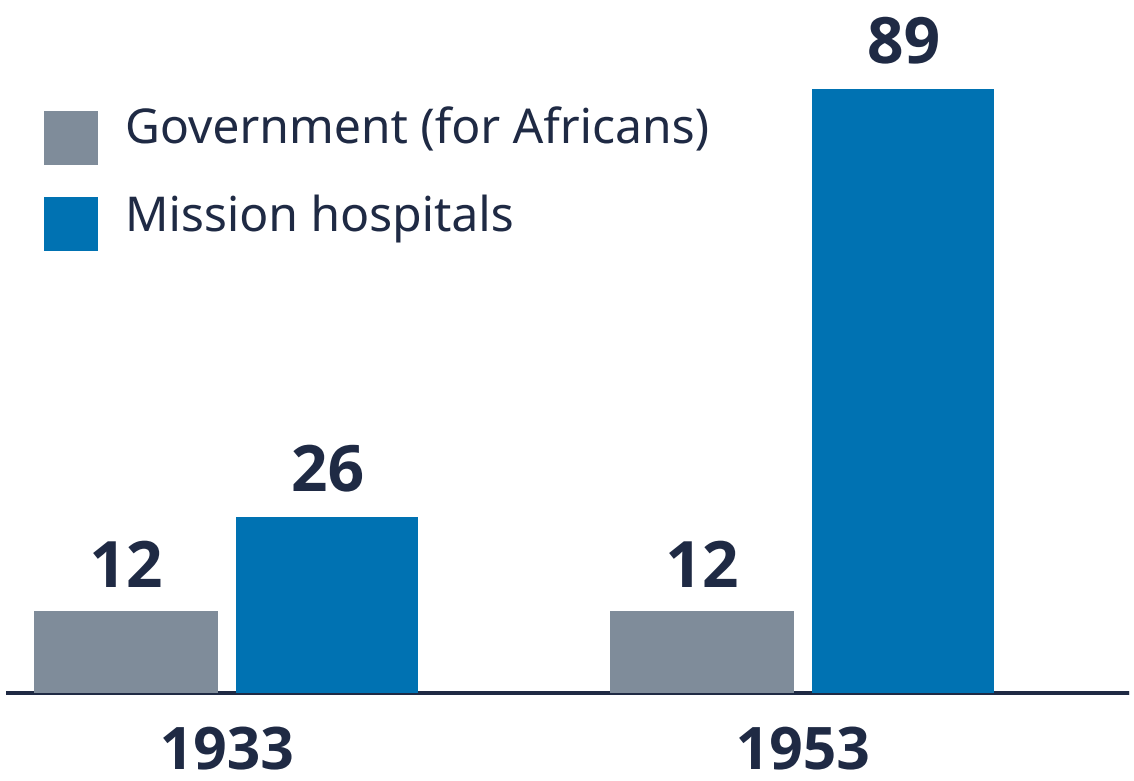
**Missing piece: why trust varies**

*[Insert map: within-country variation in facility births or DTP3 by DHS cluster, Zambia, with health facilities overlaid]*

Sources: Peters et al. 2008; Ensor and Cooper 2004; Gilson 2003; Østergaard 2015; Weiss et al. 2019

# For decades, missionaries ran most of the Western healthcare Africans met

## Hospitals in Northern Rhodesia




Source for chart: Chiseni (2025), African Studies 84(1-2), Figure 3 and text

### Trust deficit

Coercive colonial medical campaigns in central Africa still lower trust and vaccination (Lowes and Montero 2021)

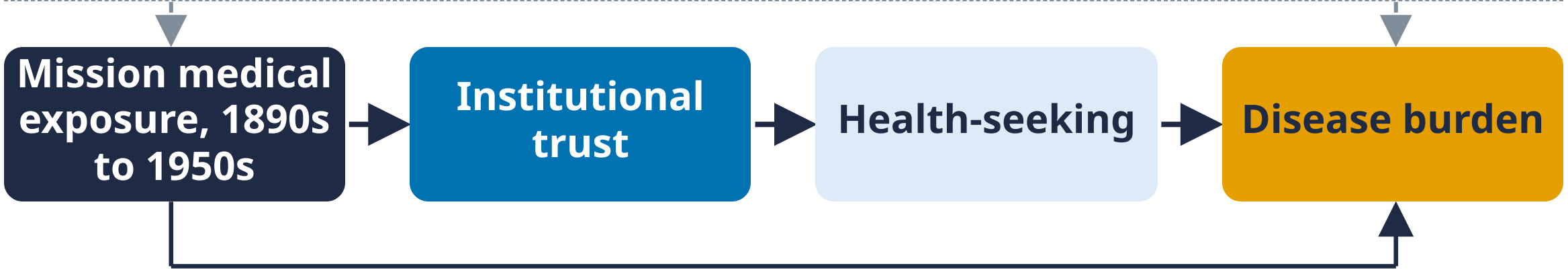
### Trust dividend

Living near a colonial medical mission in India still predicts better health (Calvi and Mantovanelli 2018)

**My hypothesis: mission medicine left a dividend in some places and a deficit in others**

# Four questions, one causal chain

I control for where missions chose to go and where the state built hospitals




Direct route: facilities that survive today

**RQ1 Map**
Where was mission care?

**RQ2 Link**
Trust, behaviour and burden today

**RQ3 Channels**
Facilities, health workers, norms, coercion

**RQ4 Meaning**
How communities see it

# I've already built this for Zambia, from records others haven't used

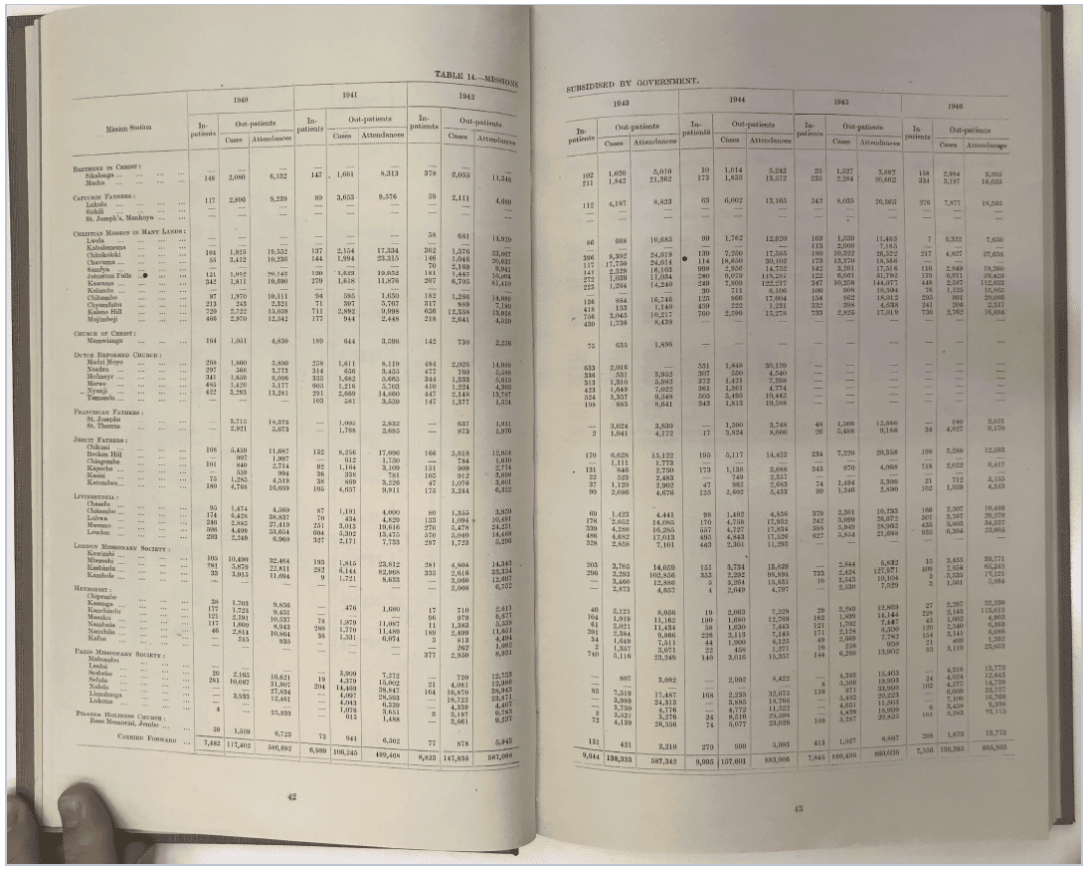


TABLE 14.—MISSIONS SUBSIDISED BY GOVERNMENT.

Northern Rhodesia medical report, Table 14: subsidised mission stations, in- and out-patients, 1940 to 1946

**76 to 91%**

of missions are missing from the atlases most studies use (Jedwab et al. 2022)

**Blue Book church returns**

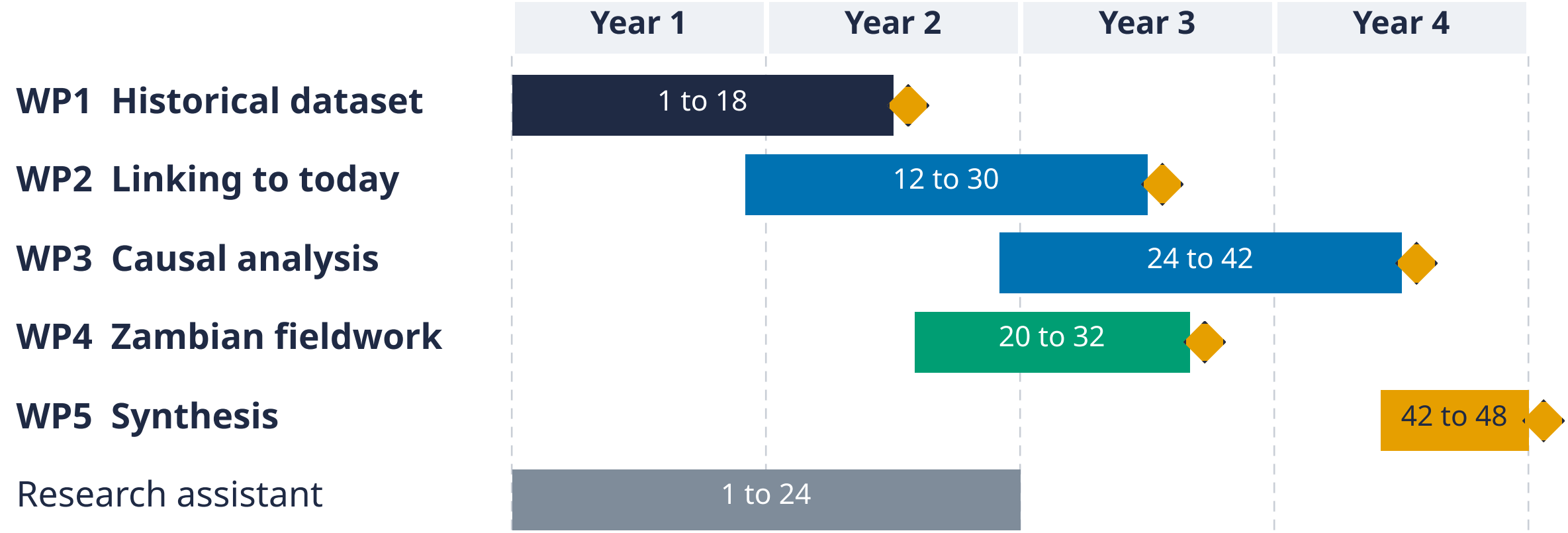
Every mission station, by society

**Colonial medical reports**

Facility type and yearly patient numbers

**= open, facility-level dataset for British Africa**

# 48 months, five work packages




**Milestones (diamonds):** v1 dataset and data paper (m18), first article (m30), fieldwork coded (m32), burden and mediation article (m42), open dataset and policy briefs (m48)

# Separating the effect of mission medicine from where missions chose to go

## The objection

Missionaries chose healthier, more accessible, richer places.

A simple comparison would mix up the effect of medicine with the effect of place.

Jedwab, Meier zu Selhausen and Moradi (2022)

1. **Control for location drivers**
   Malaria, railways, cash crops, state hospitals
2. **Medical vs non-medical stations**
   Same choice of location, different medical exposure
3. **Societies' medical propensity**
   Some societies did far more medical work than others
4. **Dose-response and distance decay**
   Effects grow with exposure, fade with distance

**Report sensitivity bounds, not a claim of perfect exogeneity**

# Asking communities what this history means

**2**
communities with long mission-hospital histories

**2**
comparison communities without

Matched on today's access to care, province and language

**Who we talk to**
Elders, health workers, healers, church and traditional leaders, young people

**~50** semi-structured interviews (40 to 60)

**8** focus groups, women and men separately

**30** days in the field with 4 Zambian assistants

**1** community feedback visit before publication

# What the reviewers asked, and what has changed

| Reviewers asked | My response |
| --- | --- |
| **How is trust measured?** | DHS blood-test refusal, Afrobarometer clinic experience, and fieldwork accounts |
| **Is the mediation credible?** | Total effect is primary; trust share via sequential g-estimation with sensitivity bounds |
| **Is continental coverage feasible?** | Coverage audit: [N] territories, [years]; core set by month 18 |

## Since submission

[Add only real updates: advisory group names, Zambian partner, coverage audit, new papers or talks]

# I've already done the hardest part once

| | |
|---|---|
| **2022** | PhD, Lund: built the Zambian dataset |
| **2024** | Missions and HIV today (EHDR) |
| **2024** | Open-access Springer monograph |
| **2024** | Gender and schooling (with J. Bolt) |
| **2025** | Mission healthcare (African Studies) |
| **Forthcoming** | IV design: missions and wealth (Elgar) |
| **Postdoc** | Six months of fieldwork in rural Zambia |

Sole author: thesis, monograph, EHDR, African Studies, Elgar chapter

## This award adds

- Scaling across many archives
- Causal mediation
- Spatial epidemiology
- Leading a team

# Stellenbosch gives me the historical-data core; partners bring the health expertise




An African host: close to field sites and policy audiences, and the skills stay on the continent

# How I'll run the team

## Train the research assistant

Written development plan; archival, GIS and data skills; co-authorship on the data paper when earned

## Fair pay and credit in Zambia

Fieldwork staff trained, paid at fair local rates, credited by name

## Communities see results first

Feedback visit to review and interpret findings before publication

## Open by default

Dataset, code and papers openly available; pre-analysis plan

*[Optional: flexible working, from my own experience of parental leave]*

# If trust is the lever, buildings alone won't close the gap

**If I'm right:** part of Africa's health inequality is historical, and trust is a lever that infrastructure alone won't move.

**Whatever the result:** the first open, facility-level map of mission medicine in British Africa.

# Thank you

Michael Chanda Chiseni | Stellenbosch University (LEAP)

# Backup slides

For questions only. Check with your Funding Manager that backup slides are allowed during Q&A.

# Backup: how I measure trust

## Revealed

Refusal of free DHS blood tests (Lowes and Montero 2021)

Same sample as the health behaviours

## Reported

Afrobarometer: ease of getting care, disrespect, informal payment

Placebo: trust in police and courts

## Lived

Fieldwork accounts of why people use, bypass or avoid clinics

**Three kinds of trust: in biomedicine, in health workers, in the health system (Gilson 2003)**

# Backup: the instrument and its limits

## Instrument

Medical propensity of the station's society, measured in other territories (leave-one-out)

## Threat

Societies also differ in doctrine and schooling, which can affect health directly

## Response

Control for denomination and schools

Balance tests across societies

Plausibly exogenous bounds (Conley, Hansen and Rossi 2012)

The instrument is one strand of evidence, not the whole case.

# Backup: mediation, stated honestly

1. Primary result: the total effect of mission medical exposure
2. Controlled direct effect via sequential g-estimation (Acharya, Blackwell and Sen 2016)
3. Sensitivity analysis: how much hidden confounding erases the trust share (Imai, Keele and Tingley 2010)
4. Common geography: DHS links are individual-level; Afrobarometer and burden at district or cluster level
5. Triangulate with the Zambian fieldwork

# Backup: risks and mitigations

| Risk | Mitigation |
| --- | --- |
| **Uneven archival coverage** | Catalogue audit before travel; richest territories first; results hold on the core set |
| **Selection of mission sites** | Layered design (slide 7); sensitivity bounds |
| **Data access (DHS, Afrobarometer)** | Download DHS rounds and GPS early; IPUMS copies; district-level fallback |
| **Approvals in Zambia** | Applications from month 1; fieldwork starts month 20 |
| **Travel or photography limits** | British Online Archives; archive copying services |

# Backup: outcomes and exposure

## Primary outcomes (DHS)

Child mortality, vaccination, antenatal care, facility births, care-seeking

Malaria test and anaemia results where collected

## Secondary (IHME)

Local Burden of Disease maps, with uncertainty carried through

Malaria: care-dependent outcomes, not prevalence

## Exposure

Distance; 10 and 25 km bands; years open; patients per head

Bands wider than DHS displacement (2 km urban, 5 km rural)

# Backup: budget at a glance

| Budget line | ZAR |
|---|---|
| **PI salary, 48 months at 100%** | 4,799,860 |
| **Research expenses (RA, fieldwork, travel, open access, equipment, other)** | 1,270,054 |
| **of which RA (24 months) / fieldwork / travel** | 325,939 / 500,190 / 182,500 |
| **Overheads (20% of direct costs)** | 1,213,983 |
| **Total requested** | 7,283,897 |